

Chapter Four:

The Theory of Production and Cost

- ❑ This chapter has **two major sections**:
- ✓ The **first part** will introduce you to the **basic concepts of production and production function, classification of inputs, essential features of short run production functions** and the **stages of short run production**.
- ✓ The **second part** mainly deals with the difference between **economic cost and accounting cost**, the **characteristics of short run cost functions**, and the **relationship between short run production functions and short run cost functions**.

4.1 Theory of production in the short run

4.1.1 Definition of production

- ✓ **Raw materials** yield **less satisfaction** to the consumer by themselves.
- ✓ For a better utility, **raw materials** **has to be transformed into outputs**.
- ✓ **However**, transforming raw materials into outputs requires inputs such as **land**, **labour**, **capital** and **entrepreneurial ability**.
- ❖ **Production** is **the process of transforming inputs into outputs**.
- ✓ The end products of the production process are **outputs** which could be **tangible (goods)** or **intangible (services)**.

4.1.2 Production function

- ❖ **Production function** is a technical relationship between **inputs** and **outputs**.
- ❖ It shows the maximum output that can be produced with fixed amount of inputs and the existing technology.

- ✓ A production function may take the form of an **algebraic equation**, **table** or **graph**.
- ✓ A general equation for production function can, for instance, be described as:

$$Q = f(X_1, X_2, X_3, \dots, X_n)$$

Where, **Q** is **output** and **$X_1, X_2, X_3, \dots, X_n$** are different **types of inputs**.

□ Inputs are commonly classified as **fixed inputs** or **variable inputs**.

❖ **Fixed inputs** are those inputs whose quantity cannot readily be changed when market conditions indicate that an immediate adjustment in output is required.

❖ In fact, **no input is ever absolutely fixed** but may be fixed during an immediate requirement.

Examples: **Buildings**, **land** and **machineries** etc.

✓ Their quantity **cannot be manipulated** easily in a short period of time.

❖ **Variable inputs** are those inputs whose quantity **can be altered almost instantaneously** in response to desired changes in output.

✓ That is, their quantities can easily be diminished when the market demand for the product decreases and vice versa.

✓ The best example of variable input is **unskilled labour**.

□ **In economics, short run** refers to a period of time in which **the quantity of at least one input is fixed**.

✓ In other words, **short run** is a time period which is **not sufficient** to **change the quantities of all inputs** so that at least one input remains fixed.

□ **Note: short run periods of different firms have different durations.**

✓ Some firms can change the quantity of all their inputs within a month while it takes more than a year for other types of firms.

- ❖ Consider a firm that uses two inputs: **capital** (fixed input) and **labour** (variable input).
- ❖ Given the assumptions of short run production, the firm can increase output only by increasing the amount of labour it uses.
- ❖ Hence, its **production function** can be given by:

$$Q = f(L)$$

where, **Q** is **output** and **L** is the **quantity of labour**.

- ❖ The **production function** shows **different levels of output** that the firm can produce by **efficiently utilizing** different units of labour and the fixed capital.
- ❖ In the above short run production function, the **quantity of capital** is **fixed**.
- ❖ **Thus**, output can change only when the amount of labour changes.

4.1.3 Total, average, and marginal product

- ✓ In production, the contribution of a variable input can be described in terms of **total**, **average** and **marginal product**.

A) Total product (TP)

- ✓ it is the *total amount of output* that can be produced by efficiently utilizing **specific combinations of the variable input and fixed input**.
- ✓ Increasing the variable input (while some other inputs are fixed) can increase the total product **only up to a certain point**.
- ✓ **Initially**, as we combine more and more units of the variable input with the fixed input, **output continues to increase**.
- ✓ **But eventually**, if we employ more and more unit of the variable input **beyond the carrying capacity of the fixed input**, output tends to decline.

- ✓ **In general, the TP function in the short-run follows a certain trend: it initially increases at an increasing rate, then increases at a decreasing rate, reaches a maximum point and eventually falls as the quantity of the variable input rises.**
- ✓ This tells us what shape a total product curve assumes.

B) Marginal Product (MP)

- ❖ it is the change in output attributed **to the addition of one unit of the variable input** to the production process, other inputs being constant.
- ❖ **For instance,** the change in total output resulting from employing additional worker (holding other inputs constant) is the marginal product of labour (**MPL**).
- ❖ In other words, **MPL** measures the **slope of the total product curve** at a given point.

$$MP_L = \frac{\Delta Q}{\Delta L} = \frac{dTP}{dL}$$

- ✓ In the **short run**, the marginal product of the variable input **first increases**, reaches its maximum and then **decreases to the extent of being negative**.
- ✓ That is, as we continue to combine more and more of the **variable input** with the **fixed input**, the marginal product of the variable input **increases initially** and **then declines**.

C) Average Product (AP)

- ✓ is the level of output that **each unit of input produces**, on the average.
- ✓ It tells us the **mean contribution of each variable input** to the total product.
- ✓ **Mathematically**, the average product of labour (APL), for instance, is given by:

$$AP_L = \frac{TP}{L}$$

- ✓ **AP_L** first increases, reaches its maximum value and **eventually declines**.

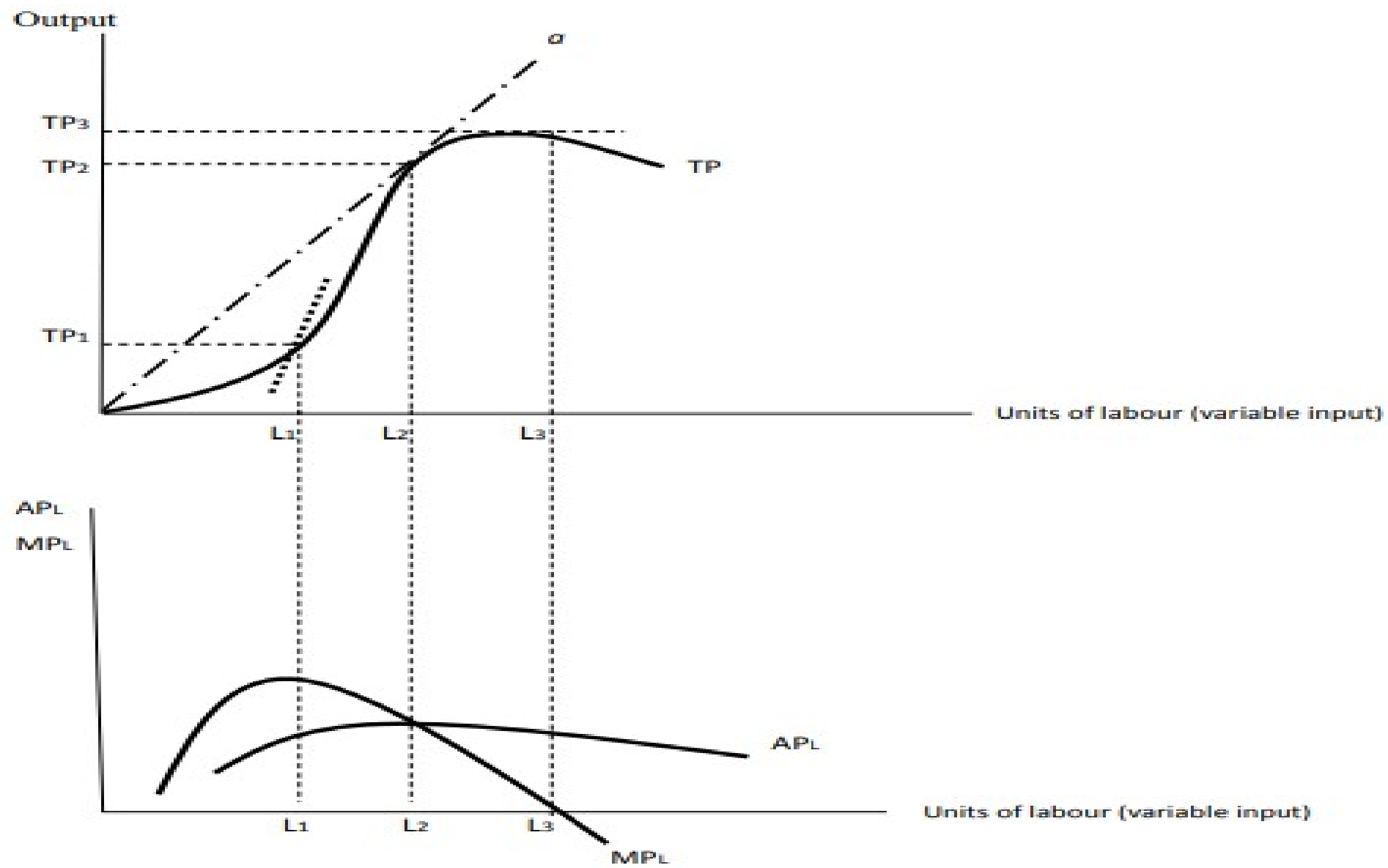


Figure 4.1: Total product, average product and marginal product curves

- ❑ The **AP curve** can be measured by the **slope of rays originating from the origin to a point on the TP curve (see figure 4.1).**
- ❖ For example, the **AP_L** at **L_2** is the ratio of **TP_2** to **L_2** = **Slope of ray a.**

- ❑ The **RELATIONSHIP** between **MP_L** and **AP_L** can be stated as follows.
 - ✓ When **APL is increasing**, **$MP_L > AP_L$.**
 - ✓ When **APL is at its maximum**, **$MP_L = AP_L$.**
 - ✓ When **APL is decreasing**, **$MP_L < AP_L$.**

Example: Suppose that the short-run production function of certain cut-flower firm is given by: **$Q = 4KL - 0.6K^2 - 0.1L^2$** , where, **Q** is quantity of cut-flower produced, **L** is labour input and K is fixed capital input (**K=5**).

- a) Determine the **AP_L** function.
- b) At what level of labour does the total output of cut-flower reach the maximum?
- c) What will be the maximum achievable amount of cut-flower production?

Solution:

$$a) AP_L = \frac{Q}{L} = \frac{4KL - 0.6K^2 - 0.1L^2}{L} = 4K - \frac{0.6K^2}{L} - 0.1L = 20 - \frac{15}{L} - 0.1L = \frac{20L - 15 - 0.1L^2}{L}$$

b) The total product level(Max Q) occurs when $MP_L = 0$.

$$MP_L = \frac{dTP}{dL} = \frac{d(20L - 15 - 0.1L^2)}{dL} = 20 - 0.2L = 0$$
$$0.2L = 20; \quad L^* = 100$$

Hence, total output will be the maximum when 100 workers are employed.

c) Substituting the optimal values of labor (L=100) and capital (K=5) into the original production function (Q):

$$Q_{max} = 4\bar{K}L^* - 0.6\bar{K}^2 - 0.1L^{*2} = 4(\bar{5})(100)^* - 0.6(\bar{5})^2 - 0.1(100)^{*2}$$
$$= 2000 - 15 - 1000 = 985$$

$$Q_{max} = 985$$

4.1.4 The law of variable proportions/LDMR

- ✓ The **law of variable proportions** states that as successive units of a variable input(say, labour) are added to a fixed input (say, capital or land), beyond some point the extra, or **marginal**, **product that can be attributed to each additional unit of the variable resource will decline**.
- ✓ **For example**, if additional workers are hired to work with a constant amount of **capital equipment**, output will **eventually rise by smaller and smaller amounts as more workers are hired**.
- Assumptions of LDMR
 - ✓ **technology is fixed** and thus the techniques of production do not change.
 - ✓ **all units of labour are assumed to be of equal quality**.
 - ✓ Each successive worker is presumed to have the **same innate ability, education, training**, and **work experience**.

- ✓ **Marginal product** ultimately diminishes **not because successive workers are less skilled or less energetic** rather it is because more workers are being used relative to the amount of plant and equipment available.
- ✓ The law starts to operate after the **marginal product curve reaches its maximum** (this happens when the number of workers exceeds L_1 in **figure 4.1**).
- ✓ This law is also called the **law of diminishing marginal returns(LDMR)**.

4.1.5 Stages of production

- ✓ We are not in a position to determine the specific number of the variable input (labour) that the firm should employ because this depends on several other factors than the productivity of labour.
- ✓ **However**, it is possible to determine the ranges **over which the variable input (labour) be employed**.
- ✓ To this end, economists have defined **three stages of short run production**.

Stage I: This stage of production covers the range of variable input levels over which the **average product (APL) continues to increase**.

- ✓ It goes from the origin to the point where the **APL is maximum**, which is the **equality of MPL and APL** (**up to L_2** level of labour employment in figure 4.1).
- ✓ This stage is **not an efficient region of production** though the MP of variable input is positive.
- ✓ **The reason** is that the variable input (the number of workers) is **too small** to efficiently run the fixed input so that **the fixed input is under-utilized (not efficiently utilized)**.

Stage II: It ranges from the point where **AP_L is at its maximum (MPL=APL)** to the point where **MPL is zero** (**from L_2 to L_3** in figure 4.1).

- ✓ Here, as the labour input increases by one unit, **output still increases but at a decreasing rate**.

- ✓ Due to this, the **second stage of production** is termed as **the stage of diminishing marginal returns.**
- ✓ The reason for decreasing average and marginal products is due to the **scarcity of the fixed factor.**
- ✓ That is, **once the optimum capital-labour combination is achieved**, employment of additional unit of the variable input will cause the output to increase at a slower rate.
- ✓ As a result, the **marginal product diminishes.**
- ✓ This stage is the **efficient region of production.**
- ✓ Additional inputs are contributing positively to the total product and MP of successive units of variable input is declining (indicating that the fixed input is **being optimally used**)._
- ✓ Hence, **the efficient region of production** is where the **marginal product of the variable input is declining but positive.**

Stage III: In this stage, **an increase in the variable input is accompanied by decline in the total product.**

- ✓ Thus, the total product curve slopes downwards, and the **marginal product of labour becomes negative.**
- ✓ This stage is also known as the **stage of negative marginal returns** to the variable input.
- ✓ The cause of negative marginal returns is the fact that **the volume of the variable inputs is quite excessive relative to the fixed input; the fixed input is over-utilized.**
- ✓ Obviously, a rational firm should not operate in **stage III** because additional units of variable input are contributing negatively to the total product (MP of the variable input is negative).
- ✓ In **figure 4.1**, this stage is indicated by the employment of **labour beyond L_3 .**

4.2 Theory of costs in the short run

4.2.1 Definition and types of costs

- ✓ To produce goods and services, firms need **factors of production** or simply **inputs**.
- ❖ To acquire these inputs, they have to buy them from resource suppliers.
- ❖ **Cost** is, therefore, the monetary value of inputs used in the production of an item.
- ❖ Economists use the term —**profit** differently from the way accountants use it.
- ❖ To the accountant, **profit** is the firm's **total revenue less its explicit costs (accounting costs)**.
- ❖ To the economist, **profit** is **total revenue less economic costs (explicit and implicit costs)**.

- ❖ **Accounting cost** is the monetary value of all purchased inputs used in production; it ignores the cost of non-purchased (self-owned) inputs.
- ❖ It considers only direct expenses such as wages/salaries, **cost of raw materials**, depreciation allowances, interest on borrowed funds and **utility expenses** (electricity, water, telephone, etc.).
- ❖ These costs are said to be explicit costs.
- ❖ **Explicit costs** are out of pocket expenses for the purchased inputs.
- ❖ If a producer calculates her cost by considering only the costs incurred for purchased inputs, then her profit will be an accounting profit.

$$\begin{aligned}\text{Accounting profit} &= \text{Total revenue} - \text{Accounting cost} \\ &= \text{Total revenue} - \text{Explicit cost}\end{aligned}$$

- ✓ In the real world economy, entrepreneurs may use some resources which may not have direct monetary expense since the entrepreneur can own these inputs himself or herself.

- ✓ **Economic cost** of producing a commodity considers the monetary value of all inputs (**purchased** and **non-purchased**).
- ✓ Calculating economic costs will be difficult since there are no direct monetary expenses for non-purchased inputs.
- ✓ The monetary value of these inputs is obtained **by estimating their opportunity costs in monetary terms**.
- ✓ The estimated **monetary cost for non-purchased** inputs is known as **implicit cost**.
- ✓ **Example:** if Mr. X quits a job which pays him Birr 10, 000.00 per month in order to run a firm he has established, then the **opportunity cost of his labour** is taken to be Birr **10,000.00** per month (the salary he has forgone in order to run his own business).
- ✓ Therefore, **economic cost** is given by the sum of **implicit cost** and **explicit cost**.

Economic profit = Total revenue – Economic cost (Explicit cost + Implicit cost)

- ✓ **Economic profit** will give the **real profit of the firm** since all costs are taken into account.
- ✓ **Accounting profit** of a firm **will be greater than economic profit** by the amount of implicit cost.
- ✓ **If all inputs are purchased from the market, accounting and economic profit will be the same.**
- ✓ However, if implicit costs exist, then accounting profit will be larger than economic profit.

4.2.2 Total, average and marginal costs in the short run

- ❖ A cost function shows the total **cost of producing a given level of output**; can be described using **equations, tables** or **curves**.
- ❖ A cost function can be represented using an equation as follows:

$$C = f(Q),$$

where **C** is the total cost of production and **Q** is the level of output.

- ❖ In **the short run**, total cost (TC) can be broken down into two – **total fixed cost (TFC)** and **total variable cost (TVC)**.
- ❖ By **fixed costs** we mean **costs which do not vary with the level of output**.
- ❖ They are regarded as fixed because these costs are **unavoidable** regardless of the level of output.
- ❖ The firm **can avoid fixed costs** only **if he/she stops operation (shuts down the business)**.
- ❖ The fixed costs may include **salaries of administrative staff**, **expenses for building depreciation** and **repairs**, expenses for **land maintenance** and the **rent of building** used for production.
- ❖ **Variable costs**, on the other hand, include all costs which **directly vary** with the level of output.
- ✓ if the firm produces zero output, the variable cost is zero.

Example: **cost of raw materials**, the **cost of direct labour** and the **running expenses of fuel**, **water**, **electricity**, etc.

- ❑ **In general**, the short run total cost is given by the sum of total fixed cost and total variable cost. That is,

$$TC = TFC + TVC$$

- ❖ Based on the definition of the short run cost functions, let's see what their shapes look like.

A) Total fixed cost (TFC): Total fixed cost is denoted by **a straight line parallel to the output axis**.

- ✓ This is because such costs do not vary with the level of output.

B) Total variable cost (TVC): The total variable cost of a firm has an **inverse S-shape**.

- ✓ The shape indicates the **law of variable proportions** in production.
- ✓ At the initial stage of production with a given plant, as more of the variable factor is employed, **its productivity increases**.
- ✓ Hence, the **TVC increases at a decreasing rate**.

- ✓ This **continues until the optimal combination of the fixed and variable factor** is reached.
- ✓ Beyond this point, as increased quantities of the variable factor are combined with the fixed factor, the productivity of the variable factor declines, and the **TVC increases at an increasing rate.**

C) Total Cost (TC): The total cost curve is obtained by **vertically adding TFC and TVC** at each level of output.

- ✓ The **shape of the TC curve follows the shape of the TVC curve**, i.e. the TC has also an **inverse S-shape.**
- ✓ It should be noted that when the level of output is zero, **TVC is also zero** which implies **$TC = TFC$.**

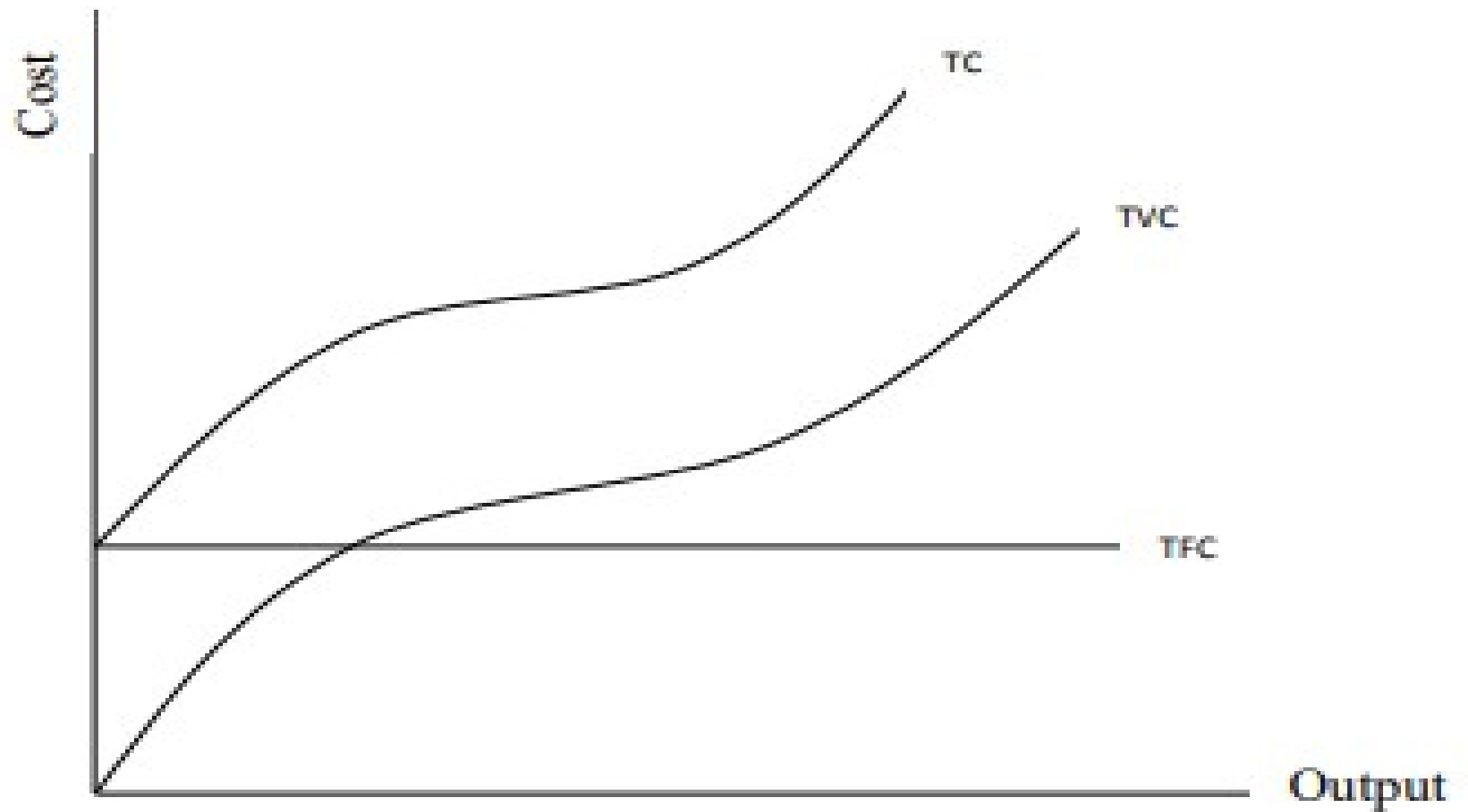


Figure 4.2: *Short run TC, TFC and TVC curves*

Per unit costs

- ✓ From total costs functions we can derive **per-unit costs**.
- ✓ These are even more **important in the short run analysis of the firm**.

a) Average Fixed Cost (AFC)

- ✓ is **total fixed cost per unit of output**.
- ✓ It is calculated by dividing TFC by the corresponding level of output.
- ✓ **The curve declines continuously** and approaches both axes asymptotically.

$$AFC = \frac{TFC}{Q}$$

b) Average Variable Cost (AVC)

- ✓ Average variable cost is **total variable cost per unit of output**.
- ✓ It is obtained by dividing total variable cost by the level of output.

$$AVC = \frac{TVC}{Q}$$

- ✓ The short run **AVC** falls initially, **reaches its minimum**, and then **starts to increase**.
- ✓ Hence, the AVC curve has **U-shape** and the reason behind is the **law of variable proportions**.

c) Average total cost (ATC) or (AC)

- ✓ **is the total cost per unit of output.**
- ✓ It is calculated by dividing the total cost by the level of output.

$$AC = \frac{TC}{Q}$$

- ✓ Equivalently, **$AVC = \frac{TFC+TVC}{Q} = AFC + AVC$**
- ✓ Thus, AC can also be given by **the vertical sum of AVC and AFC**.

D) Marginal Cost (MC)

- ❖ Marginal cost is defined as the additional cost that a firm incurs to produce one extra unit of output.
- ❖ In other words, it is the change in total cost which results from a unit change in output.
- ❖ Graphically, MC is the slope of TC function.

$$MC = \frac{dT C}{dQ}$$

- ❖ In fact, MC is also a change in TVC with respect to a unit change in the level of output.

$$MC = \frac{dTVC + dTFC}{dQ} = \frac{dTVC}{dQ}, \quad \text{since } \frac{dTFC}{dQ} = 0$$

- ❖ Given inverse S-shaped TC and TVC curves, MC initially decreases, reaches its minimum and then starts to rise.
- ❖ From this, we can infer that the reason for the MC to exhibit U-shape is also the law of variable proportions.

❖ In summary, **AVC**, **AC** and **MC** curves are all U-shaped due to the law of variable proportions.

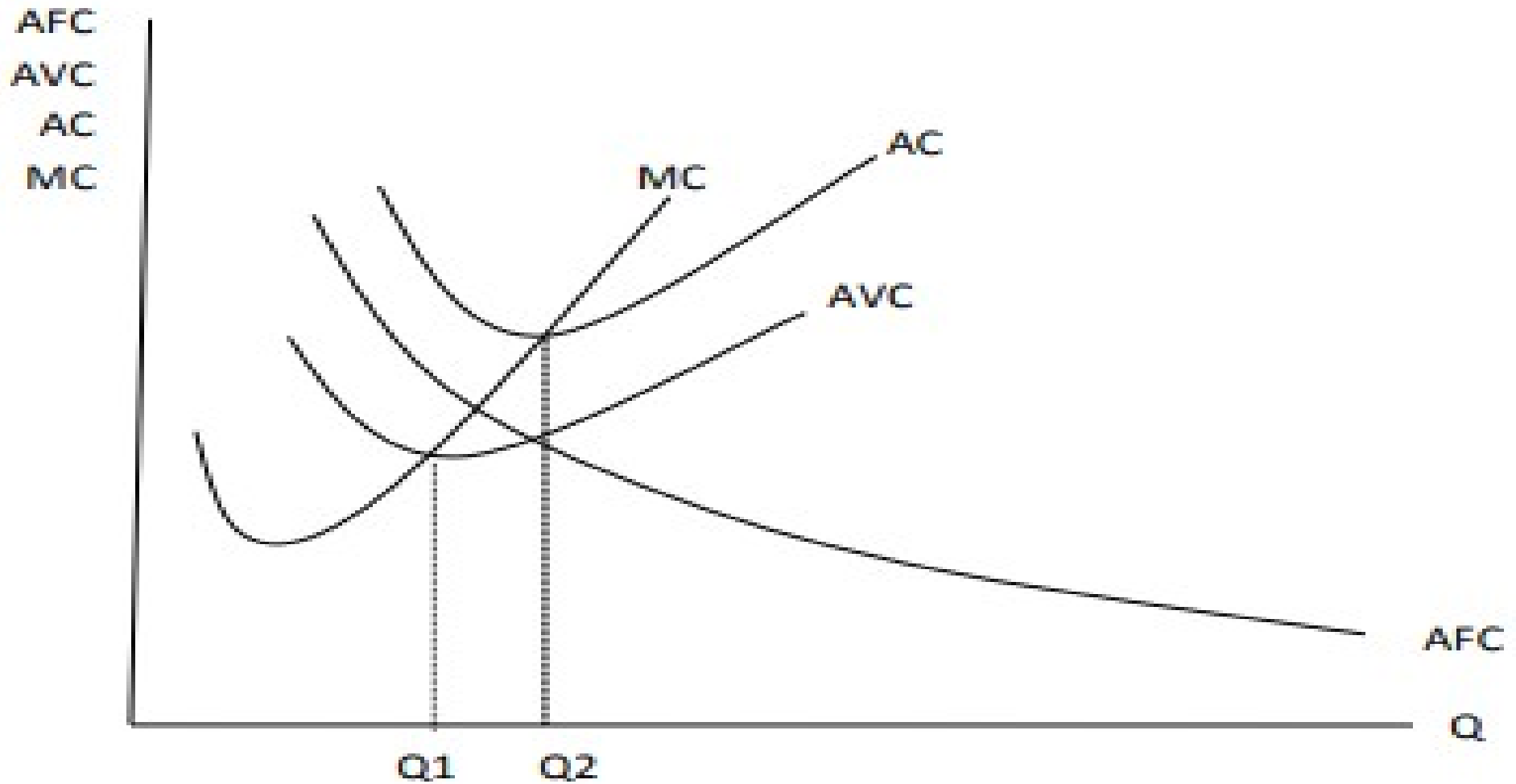


Figure 4.3: Short run AFC, AVC, AC and MC Curves

- ✓ In the above figure, **the AVC curve reaches its minimum point at Q_1** level of output and **AC reaches its minimum point at Q_2** level of output.
- ✓ The **vertical distance between AC and AVC**, that is, **AFC** decreases continuously as output increases.
- ✓ It can also be noted that the **MC curve passes through the minimum points of both AVC and AC curves.**

Example: Suppose the short run cost function of a firm is given by:

$$TC = 2Q^3 - 2Q^2 + Q + 10$$

- Find the expression of **TFC** & **TVC**
- Derive the expressions of **AFC**, **AVC**, **AC** and **MC**
- Find the levels of output that minimize **MC** and **AVC** and then find the minimum values of **MC** and **AVC**

Solution:

Given $2Q^3 - 2Q^2 + Q + 10$

A) $TFC = 10$; and $TVC = 2Q^3 - 2Q^2 + Q$

B) $AFC = \frac{TFC}{Q} = \frac{10}{Q}$

$$AVC = \frac{TVC}{Q} = \frac{2Q^3 - 2Q^2 + Q}{Q} = 2Q^2 - 2Q + 1$$

$$AC = \frac{TC}{Q} = \frac{2Q^3 - 2Q^2 + Q + 10}{Q} = 2Q^2 - 2Q + 1 + \frac{10}{Q}$$

$$MC = \frac{dTC}{dQ} = \frac{d(2Q^3 - 2Q^2 + Q + 10)}{dQ} = 6Q^2 - 4Q + 1$$

C) To find the minimum value of MC, $\frac{dMC}{dQ} = 0 = 12Q - 4 = 0$

$$Q = 1/3$$

✓ Thus, MC is minimized when $Q = 0.33$

- ✓ The **minimum value of MC** will be:

$$MC = 6Q^2 - 4Q + 1 = 6\left(\frac{1}{3}\right)^2 - 4\left(\frac{1}{3}\right) + 1 = 0.33$$

- To find the **minimum value of AVC**, $\frac{dAVC}{dQ} = 0 = 4Q - 2 = 0$

$$Q = 0.5$$

- ✓ AVC is minimized at **$Q = 0.5$** .
✓ The **minimum value of AVC** will be:

□ $AVC = 2Q^2 - 2Q + 1$

$$AVC = 2(0.5)^2 - 2(0.5) + 1 = 0.5 - 1 + 1 = 0.5$$

4.2.3 The relationship between short run production and cost curves

- ❖ Suppose a firm in the **short run** uses labour as a variable input and capital as a fixed input.
- ❖ Let the price of labour be given by **w**, which is constant.
- ❖ Given these conditions, we can derive the relation between **MC** and **MPL** as well as the relation between **AVC** and **APL**.

i) MC and MP of Labour

$$\mathbf{MC} = \frac{dT\mathbf{C}}{dQ} = \frac{\Delta T\mathbf{VC}}{\Delta Q}, \text{ Where } T\mathbf{VC} = w * L$$

$$\text{Thus, } \mathbf{MC} = \frac{\Delta(w*L)}{\Delta Q} = w \frac{\Delta L}{\Delta Q}; \text{ but } \frac{\Delta L}{\Delta Q} = \frac{1}{MP_L}$$

➤ Therefore, $\mathbf{MC} = \frac{w}{MP_L}$

- ✓ The expression $MC = \frac{w}{MP_L}$ shows that MC and MP_L are inversely related.
- ✓ When initially MPL increases, **MC decreases**; when MPL is at its maximum, MC must be at a minimum and when finally MPL declines, MC increases.

II) AVC and AP of Labour

$$AVC = \frac{TVC}{Q}, \text{ Where } TVC = w * L$$

$$\text{Thus, } AVC = \frac{(w*L)}{Q} = w \frac{L}{Q}; \text{ but } \frac{L}{Q} = \frac{1}{AP_L}$$

$$\text{Therefore, } AVC = \frac{w}{AP_L}$$

- ✓ This expression also shows **inverse relation between AVC and APL**.
- ✓ When APL increases, AVC decreases; when APL is at a maximum, AVC is at a minimum and when finally APL declines, AVC increases.

✓ Graphically,

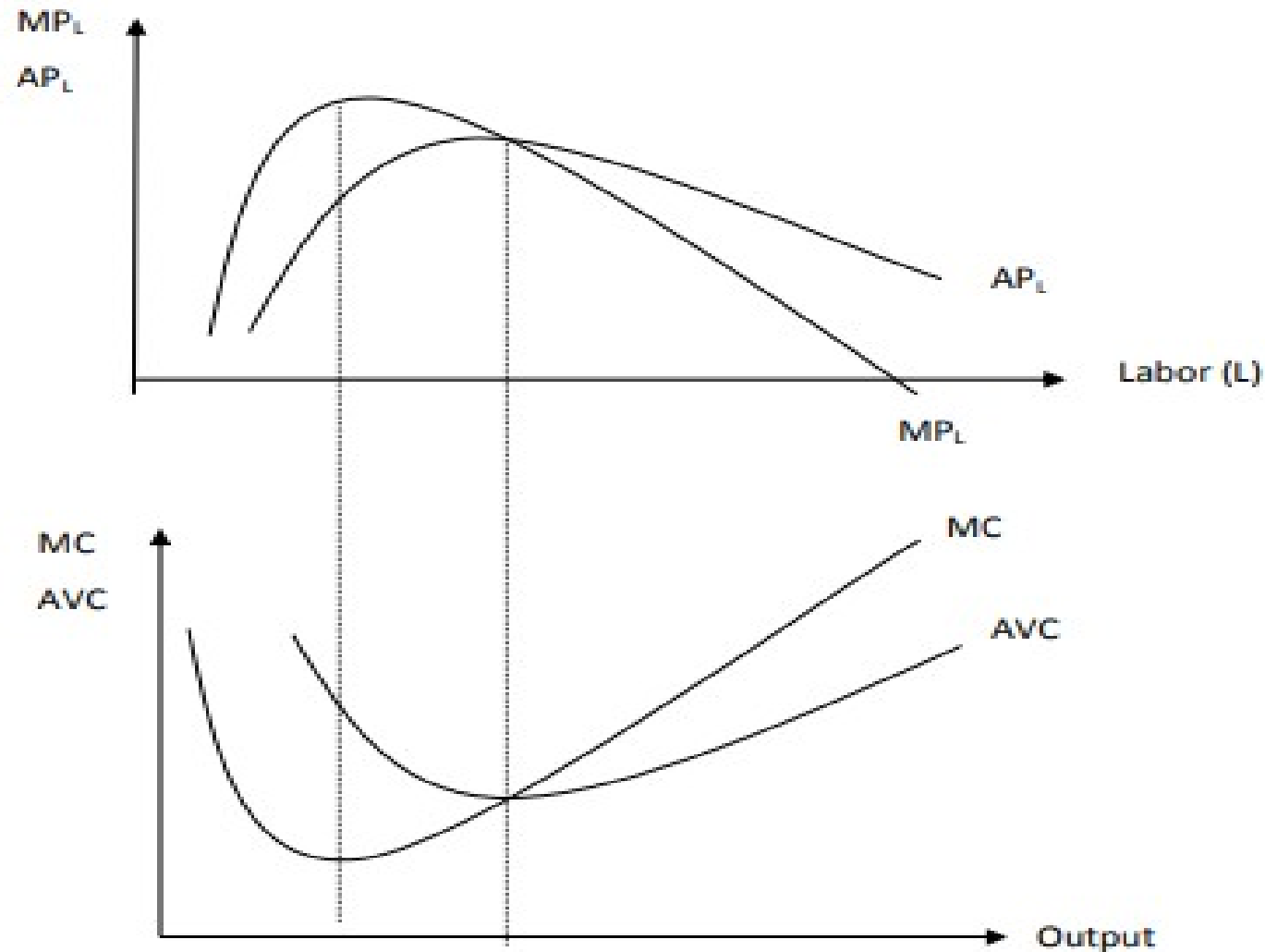


Figure 4.4: relationship between short run production and cost curves

- From the above figure, we can conclude that the **MC** curve is the mirror image of **MPL** curve and **AVC** curve is the mirror image of **APL** curve.